Linear Graphs: Plotting $y = mx + c$

Grade 5 Mathematics • The Magic Rule of Straight Lines

🛺 Meet Rohan's Auto Ride!

Rohan takes an auto-rickshaw to the cricket ground. The driver charges a fixed **base fare of ₹23** just to start the meter, plus **₹10 for every kilometer** traveled!

Formula: $Fare = Base\ Fare + (Rate \times Distance)$ ➔ $y = 10x + 23$

🗺️ 1. The Coordinate Map

A coordinate plane is like a treasure map made of two intersecting lines:

- **x-axis:** Goes horizontally (left to right)
- **y-axis:** Goes vertically (bottom to top)
- **Origin $(0,0)$:** Where the two lines cross!

Every point has an address: (x, y).

✨ 2. The Magic Equation

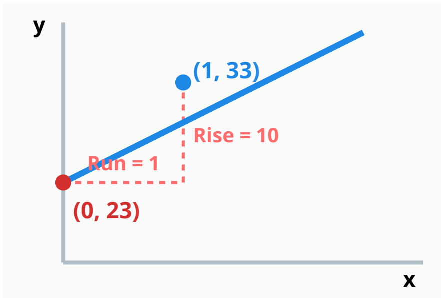
$$y = mx + c$$

- x = The input number (e.g., Distance)
- y = The output answer (e.g., Total Fare)
- m = **Steepness** (rate of change / step size)
- c = **Starting Value** (where line touches y-axis)

📉 3. Understanding 'c' and 'm'

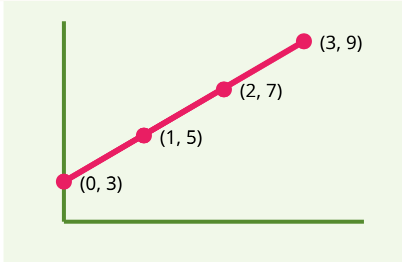
Starting Value ($c = 23$): Before moving ($x = 0$), the meter starts at 23 on the y-axis.

Steepness ($m = 10$): For every 1 km traveled ($Run = 1$), the fare rises by ₹10 ($Rise = 10$).



📍 5. Plotting $y = 2x + 3$

Plot points and connect them with a straight line!



🔢 4. Let's Calculate: $y = 2x + 1$

Find y by substituting values of x:

Input (x)	Calculation ($2x + 1$)	Output (y)	Point (x, y)
0	$2(0) + 1$	**1**	(0, 1)
1	$2(1) + 1$	**3**	(1, 3)
2	$2(2) + 1$	**5**	(2, 5)

⚡ Interactive Challenge: Predict the Y-Value!

If $y = 3x + 2$, what is y when $x = 2$?

A) 6　B) 8 ✓　C) 10　D) 12

Explanation: $y = 3(2) + 2 = 6 + 2 = 8$. Great job!

💡 Quick Summary:

- x = Input | y = Output
- m = Steepness or step size (how fast y grows)
- c = Starting value on the y-axis (when $x = 0$)
- Calculate coordinates ➔ Plot points on the grid ➔ Connect to form a straight line!